

7. THERMODYNAMIC PROPERTY RELATIONS

Introduction:

- In thermodynamics, property relations are mathematical expressions that relate thermodynamic properties such as pressure (P), temperature (T), specific volume (v), internal energy (u), enthalpy (h), and entropy (s).
- These relations allow us to determine changes in properties during a process when direct experimental data are unavailable.
- Some thermodynamic properties can be measured directly, but many others cannot.
- Therefore, it is necessary to develop some relations between these two groups so that the properties that cannot be measured directly can be evaluated.
- The derivations are based on the fact that properties are **point functions**, and the state of a simple, compressible system is completely specified by any **two independent, intensive properties**.

Some Mathematical Preliminaries

Thermodynamic properties are continuous point functions and have exact differentials.

A property of a single component system may be written as general mathematical function $z = z(x,y)$.

Note: x, y, z are independent intensive properties.

The total differential of z is written as:

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

This differential can also expressed as

$$dz = M(x,y)dx + N(x,y)dy$$

Where

$$M(x,y)dx = \left(\frac{\partial z}{\partial x}\right)_y dx$$

And

$$N(x,y)dy = \left(\frac{\partial z}{\partial y}\right)_x dy$$

Taking the partial derivative of M with respect to y and of N with respect to x yields

$$\left(\frac{\partial M}{\partial y}\right)_x = \frac{\partial^2 z}{\partial x \partial y} \text{ and } \left(\frac{\partial N}{\partial x}\right)_y = \frac{\partial^2 z}{\partial y \partial x}$$

The order of differentiation is immaterial for properties since they are continuous point functions and have exact differentials.

Therefore, the two relations above are identical:

$$\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

Note: Cyclic Relation:

$$\left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial x}\right)_y = 1 \rightarrow \left(\frac{\partial x}{\partial z}\right)_y = \frac{1}{(\partial z / \partial x)_y}$$

$$\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z = -\left(\frac{\partial z}{\partial y}\right)_x \rightarrow \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$

The first relation is called the **reciprocity relation**, and it shows that the inverse of a partial derivative is equal to its reciprocal.

The second relation is called the **cyclic relation**, and it is frequently used in thermodynamics.

THE MAXWELL RELATIONS

The equations that relate the partial derivatives of properties P , v , T , and s of a simple compressible system to each other are called the *Maxwell relations*.

They are obtained from the four Gibbs equations by exploiting the exactness of the differentials of thermodynamic properties.

Fundamental property relations of pure substance in a closed system:

Internal Energy

1st law Thermodynamics, $du = dq - dw$ ----- (1)

Definition of entropy, $ds = \frac{dq}{T} \rightarrow dq = Tds$ ----- (2)

$$Pdv \text{ work, } dw = pdv \text{ ----- (3)}$$

From (2) and (3) , eqn (1) becomes, $du = Tds - pdv \text{ ----- (4)}$

$$u = u(s,v)$$

Enthalpy

$$h = u + pv$$

$$dh = du + d(PV) = du + pdv + vdp$$

From eqn (4) $dh = Tds - Pdv + Pdv + vdp$

$$dh = Tds + vdp \text{ ----- (5)}$$

$$h=h(s,p)$$

- At this point, we we introduce the spefic Helmholtz function a and the specific Gibbs function g.
- These are defined by

$$A = u - Ts \text{ -----6}$$

And

$$g = h - Ts \text{ -----7}$$

- These functions are useful in the thermodynamics of chemical reactions.
- Differentiating equation (5) and (6)

$$da = du - T ds - s dT$$

$$dg = dh - T ds - s dT$$

Simplifying the preceding relations by using equation (4) and equation (5)

$$da = -s dT - P dv \text{ ----- (8)}$$

$$a = a(T, v)$$

$$dg = -s dT + v dP \text{ ----- (9)}$$

$$g = g(T, p)$$

A careful examination of the four Gibbs relations reveals that they are of the form

$$dz = M dx + N dy \text{ -----(10)}$$

With

$$\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y \text{ -----(11)}$$

Derivation of Maxwell Relation

a) From internal energy $U(S, V)$

$$dU = T dS - P dV$$

$$\left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial P}{\partial S} \right)_V$$

b) From Enthalpy $H(S, P)$

$$dH = T dS + V dP$$

$$\left(\frac{\partial T}{\partial P} \right)_S = \left(\frac{\partial V}{\partial S} \right)_P$$

c) From Helmholtz Free Energy $A(T, V)$

$$dA = -S dT - P dV$$

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V$$

d) From Gibbs Free Energy $G(T, P)$

$$dG = -S dT + V dP$$

$$\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P$$

- since u , h , a , and g are properties and thus have exact differentials. Applying Eq. 11 to each of them, we obtain

$$\left(\frac{\partial T}{\partial u}\right)_s = -\left(\frac{\partial P}{\partial s}\right)_u \quad \text{-----} 1$$

$$\left(\frac{\partial T}{\partial P}\right)_s = \left(\frac{\partial u}{\partial s}\right)_P \quad \text{-----} 2$$

$$\left(\frac{\partial s}{\partial u}\right)_T = \left(\frac{\partial P}{\partial T}\right)_u \quad \text{-----} 3$$

$$\left(\frac{\partial s}{\partial P}\right)_T = -\left(\frac{\partial u}{\partial T}\right)_P \quad \text{-----} 4$$

These are called the Maxwell relations.

Example

Verify the last Maxwell above relation above for steam at 250°C and 300 kpa.

Solution

Using property value from the tables (Table 6 in this case) at or about the specified state

$$\left(\frac{\partial s}{\partial p}\right)_T = -\left(\frac{\partial v}{\partial T}\right)_p$$

$$\left(\frac{\Delta p}{\Delta p}\right)_{T=250^\circ\text{C}} \cong \left(\frac{\Delta v}{\Delta T}\right)_{p=300 \text{ kpa}}$$

$$\left[\frac{s_{400\text{kpa}} - s_{200\text{kpa}}}{(400 - 200)\text{kpa}}\right]_{T=250^\circ\text{C}} \cong \left[\frac{v_{300\text{kpa}} - v_{200^\circ\text{C}}}{(300 - 200)\text{k}}\right]_{T=300\text{kpa}}$$

$$\frac{(7.3804 - 7.7100)^{\text{kJ/kg.K}}}{(400 - 200)\text{kpa}} \cong \frac{(0.87535 - 0.71643)^{\text{m}^3/\text{kg}}}{(300 - 200)^\circ\text{C}}$$

$$-0.00165 \text{ m}^3/\text{kg}\cdot\text{K} \cong -0.00159 \text{ m}^3/\text{kg}\cdot\text{K}$$

since $\text{kJ} = \text{kPa}\cdot\text{m}^3$ and $\text{K} \equiv ^\circ\text{C}$ for temperature differences. The two values are within 4 percent of each other. This difference is due to replacing the differential quantities with relatively large finite quantities

Discussion This example shows that the entropy change of a simple compressible system during an isothermal process can be determined from a knowledge of the easily measurable properties P , v , and T alone.